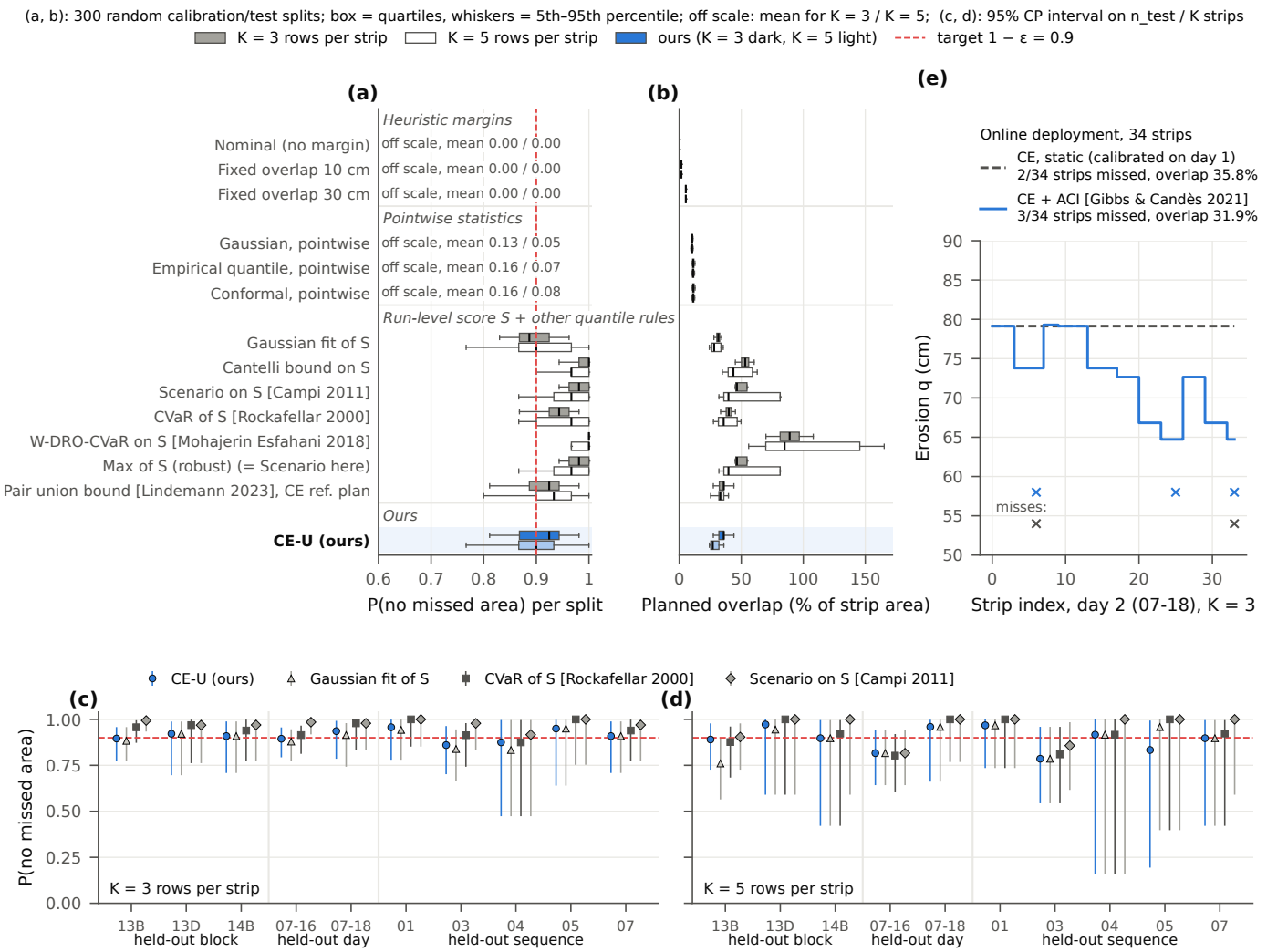
(a, b): 300 random calibration/test splits; box = quartiles, whiskers = 5th–95th percentile; off scale: mean for K = 3 / K = 5; (c, d): 95% CP interval on n_test / K strips
K = 3 rows per strip
K = 5 rows per strip
ours (K = 3 dark, K = 5 light)
target 1 − ε = 0.9
(a)
(b)
(e)
Heuristic margins
Nominal (no margin)
Fixed overlap 10 cm
Fixed overlap 30 cm
off scale, mean 0.00 / 0.00
Pointwise statistics
Gaussian, pointwise
Empirical quantile, pointwise
Conformal, pointwise
off scale, mean 0.13 / 0.05
off scale, mean 0.16 / 0.07
off scale, mean 0.16 / 0.08
Run-level score S + other quantile rules
Gaussian fit of S
Cantelli bound on S
Scenario on S [Campi 2011]
CVaR of S [Rockafellar 2000]
W-DRO-CVaR on S [Mohajerin Esfahani 2018]
Max of S (robust) (= Scenario here)
Pair union bound [Lindemann 2023], CE ref. plan
Ours
CE-U (ours)
0.6
0.7
0.8
0.9
1
P(no missed area) per split
0
50
100
150
Planned overlap (% of strip area)
Online deployment, 34 strips
CE, static (calibrated on day 1)
2/34 strips missed, overlap 35.8%
CE + ACI [Gibbs & Candès 2021]
3/34 strips missed, overlap 31.9%
Erosion q (cm)
90
85
80
75
70
65
60
55
50
misses:
0
10
20
30
Strip index, day 2 (07-18), K = 3
CE-U (ours)
Gaussian fit of S
CVaR of S [Rockafellar 2000]
Scenario on S [Campi 2011]
(c)
(d)
P(no missed area)
1.00
0.75
0.50
0.25
0.00
K = 3 rows per strip
K = 5 rows per strip
13B
13D
14B
07-16
07-18
01
03
04
05
07
held-out block
held-out day
held-out sequence